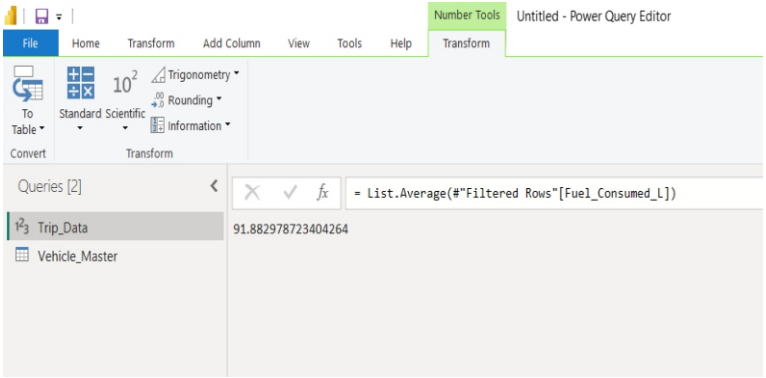
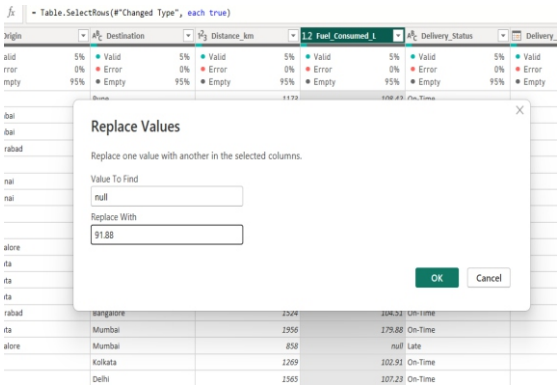
AI-POWERED FLEET PERFORMANCE & DELIVERY EFFICIENCY DASHBOARD

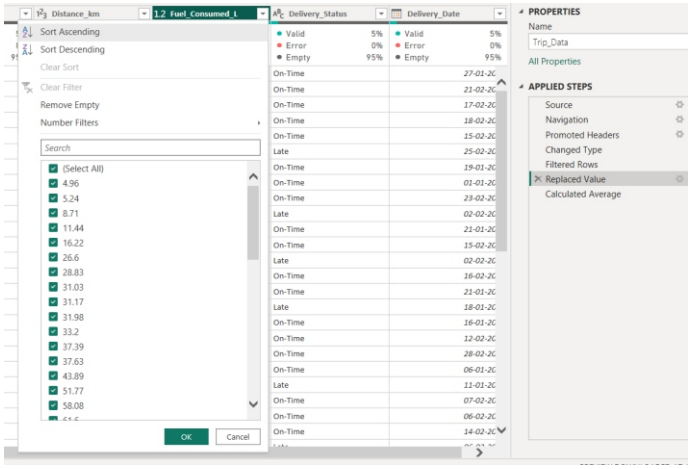
- This project focuses on building an AI-powered transport operations dashboard to analyze fleet performance and delivery efficiency. The dashboard was created using operational trip data and vehicle master data to provide meaningful insights for optimizing routes, reducing costs, and improving fleet utilization.
- The dashboard includes key performance indicators such as average delivery time and average cost per kilometer, which give a quick overview of delivery effort and transportation cost efficiency. These metrics help understand overall operational performance at a glance.
- To evaluate delivery reliability, an On-Time Delivery % by Destination bar chart was created. This visualization highlights destinations with higher or lower on-time delivery rates, enabling better route planning and corrective actions for delay-prone locations.
- Fuel usage trends were analyzed using a Fuel Efficiency by Delivery Date line chart. This chart shows how fuel efficiency changes over time and helps identify periods of higher fuel consumption or inefficiency.
- Vehicle-related costs were analyzed using a Vehicle Type vs Average Maintenance Cost pie chart. This visual compares maintenance costs across different vehicle types and supports informed decisions on fleet composition and maintenance planning.
- In addition to standard visuals, AI-powered visuals were incorporated to enhance analytical capabilities. A Q&A visual allows users to ask natural language questions and receive instant insights.The Key Influencers visual identifies factors affecting delivery status, such as distance, vehicle type, and driver. The Decomposition Tree breaks down cost per kilometer to identify major cost drivers.
- An interactive vehicle type slicer was added to enable focused analysis across all visuals. Overall, this dashboard provides actionable insights to optimize delivery routes, control operational costs, and improve fleet usage, fulfilling the expected outcome of an AI-driven transport analytics solution.

Data Cleaning –

Fuel Consumption Handling Missing values were identified in the Fuel_Consumed_L column.To ensure accurate calculations, the average fuel consumption value was calculated using Power Query. All null fuel values were then replaced with this calculated average.This step helped maintain data consistency and avoided incorrect results in further analysis.

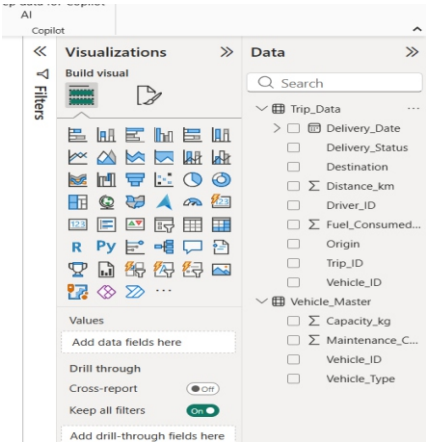






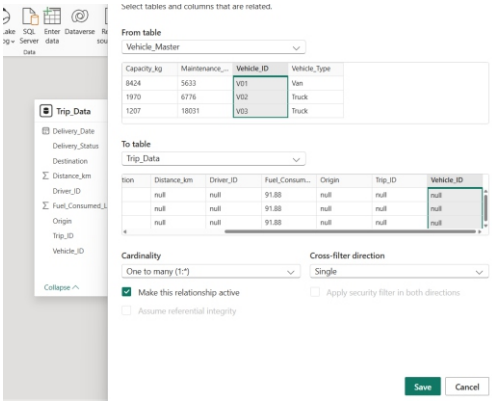
Data Preparation –

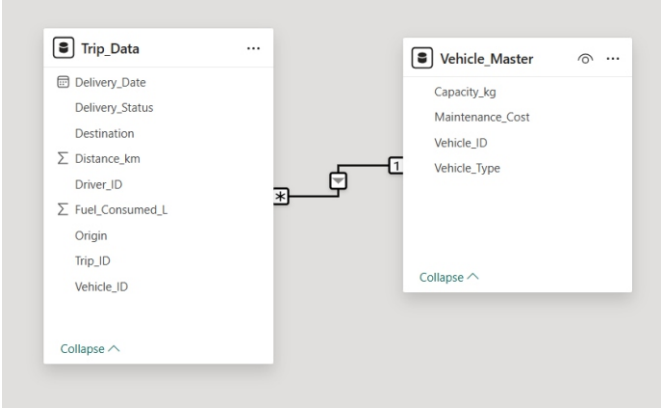
Power Query StepsThe dataset was cleaned using Power Query Editor.Unnecessary blank rows were reviewed and ignored to prevent data loss.Data types were validated, and required transformations were applied before loading the data.Finally, Close & Apply was used to load the cleaned data into the Power BI model.



Data Modeling –

Relationship CreationA relationship was created between Trip_Data and Vehicle_Master tables using Vehicle_ID. Duplicate and blank entries were removed from the master table to ensure uniqueness. A One-to-Many (1:*) relationship was established to support accurate filtering and analysis.This data model enables efficient analysis of trip performance by vehicle attributes.



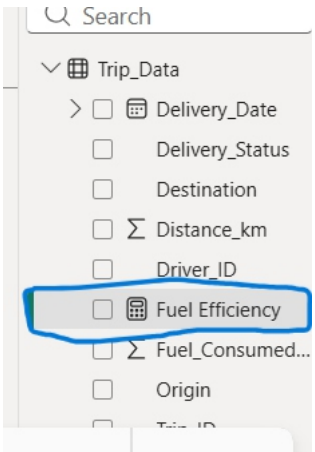


Measure 1 :

Fuel Efficiency Fuel Efficiency measure was created to evaluate vehicle performance. It calculates the total distance travelled divided by the total fuel consumed. This measure helps in understanding how efficiently fuel is utilized across all trips. Higher fuel efficiency indicates better vehicle and route performance.

Structure | Formatting | Properties

```
Fuel Efficiency = SUM(Trip_Data[Distance_km])/SUM(Trip_Data[Fuel_Consumed_L])
```

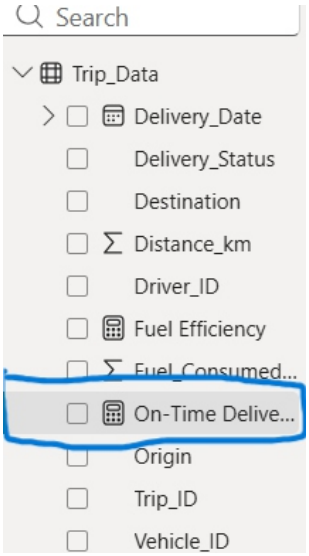


Measure 2 :

On-Time Delivery %On-Time Delivery % measure was created to analyze delivery punctuality.It calculates the percentage of trips delivered on time compared to total trips.This measure helps in evaluating overall delivery reliability and service quality.A higher percentage indicates better operational efficiency.

Structure | Formatting | Properties | Calculations

```
On-Time Delivery % = DIVIDE(CALCULATE(COUNTROWS(Trip_Data),Trip_Data[Delivery_Status]="On-Time"),COUNTROWS(Trip_Data))
```

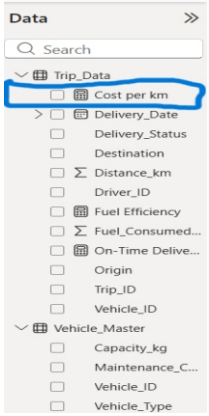


Measure 3 :

Cost per kmCost per km measure was created to analyze transportation cost efficiency. It divides the total operational cost (fuel cost and maintenance cost) by the total distance travelled. This measure helps in identifying how much cost is incurred per kilometer. Lower cost per kilometer indicates better cost management.

Structure | Formatting | Properties | Calculation

```
Cost per km = DIVIDE(100+SUM(Vehicle_Master[Maintenance_Cost]),SUM(Trip_Data[Distance_km]))
```

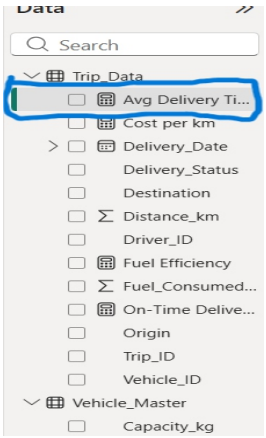


Measure 4 :

Average delivery time was approximated using average trip distance due to the absence of delivery duration data.

Structure | Formatting

```
Avg Delivery Time = AVERAGE(Trip_Data[Distance_km])
```

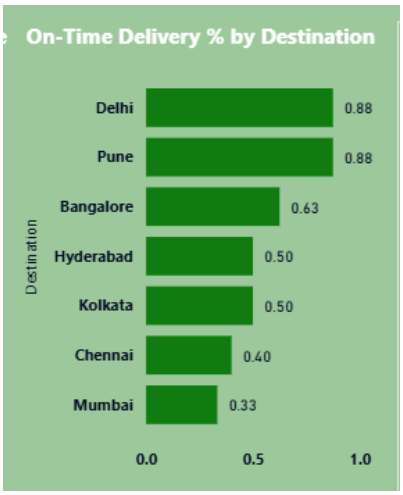


Card 1 : Avg. Delivery Time

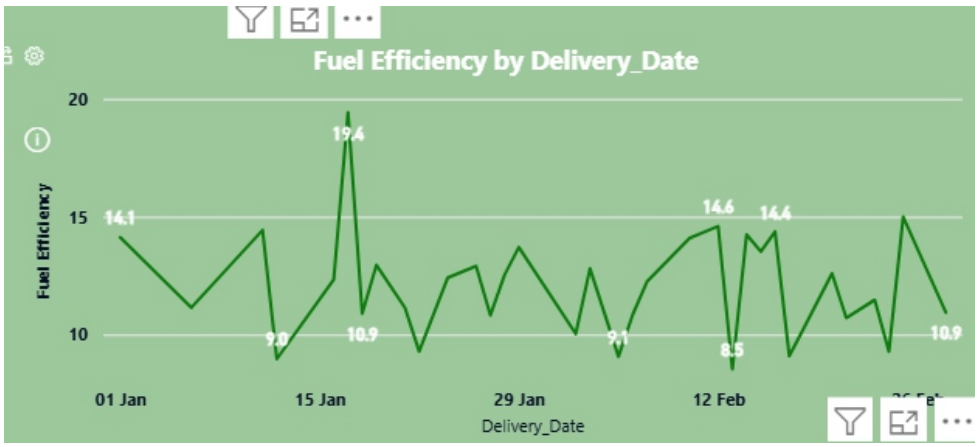
A card visual was used to display the average delivery time for all trips. Since delivery duration data was not available, average trip distance was used as a proxy to represent delivery time. This card helps in understanding the overall delivery effort required across trips.

Card 2: Average Cost per km

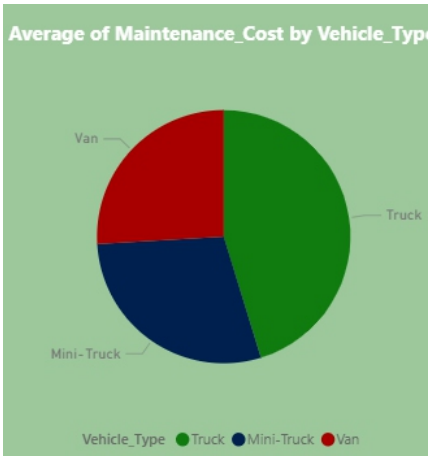
A card visual was used to display the average cost per kilometer.This value is calculated using fuel cost and maintenance cost divided by total distance travelled.The card provides a quick view of transportation cost efficiency.



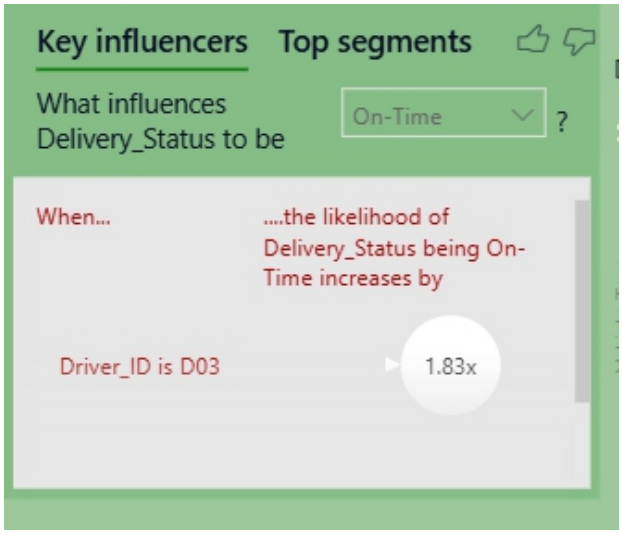
A bar chart was created to compare on-time delivery performance across different destinations.



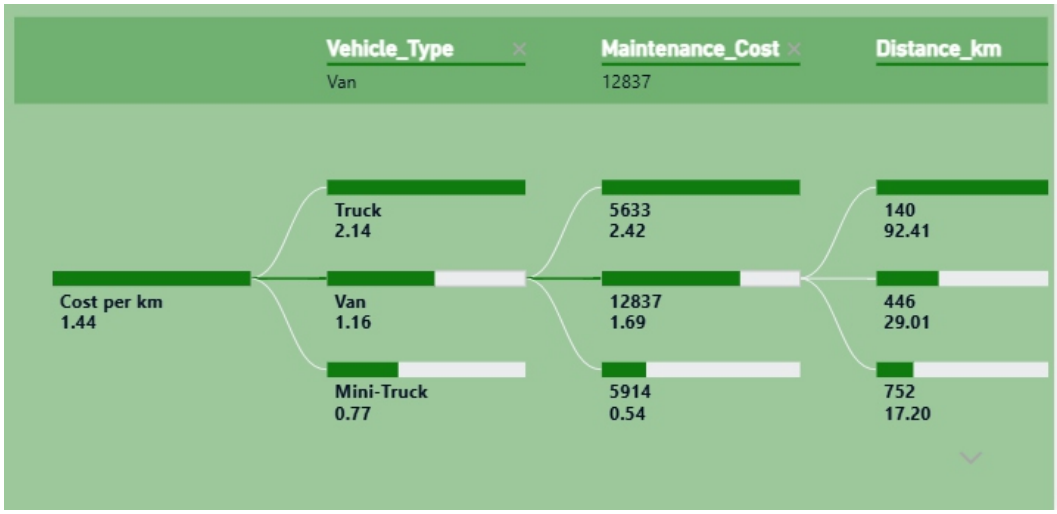
The line chart itself represents the fuel efficiency trend over time, so no additional trend line was required.



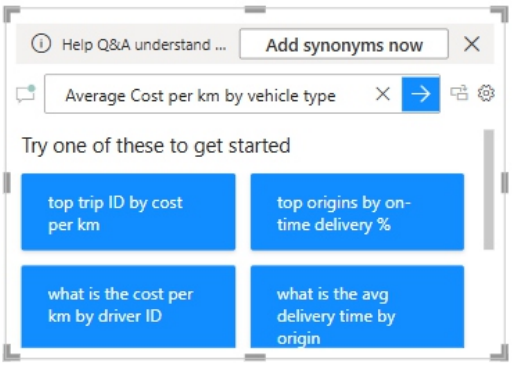
A pie chart was used to compare the average maintenance cost across different vehicle types.

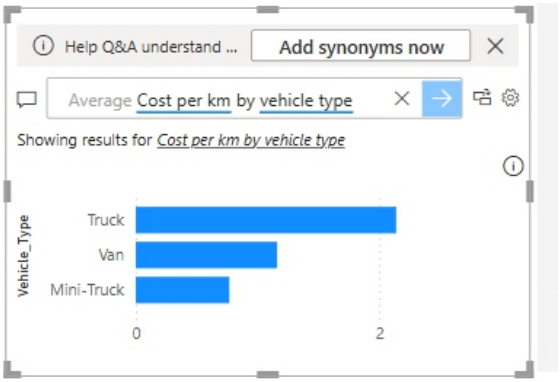


A key influencers visual was used to identify the factors influencing delivery status, revealing how driver ID, distance, and vehicle type impact on-time deliveries.

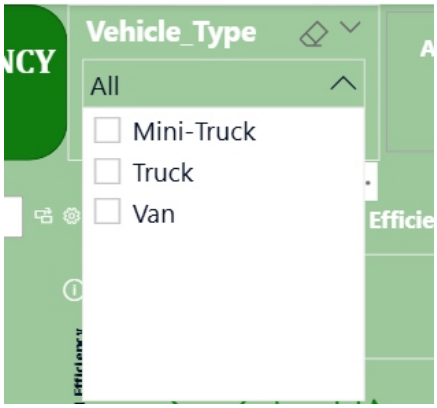


Decomposition Tree (AI Visual)A decomposition tree visual was used to analyze cost per kilometer by breaking it down by vehicle type, maintenance cost, and distance traveled. This visual helps identify the key cost drivers and supports data-driven decisions to optimize fleet usage and route planning.

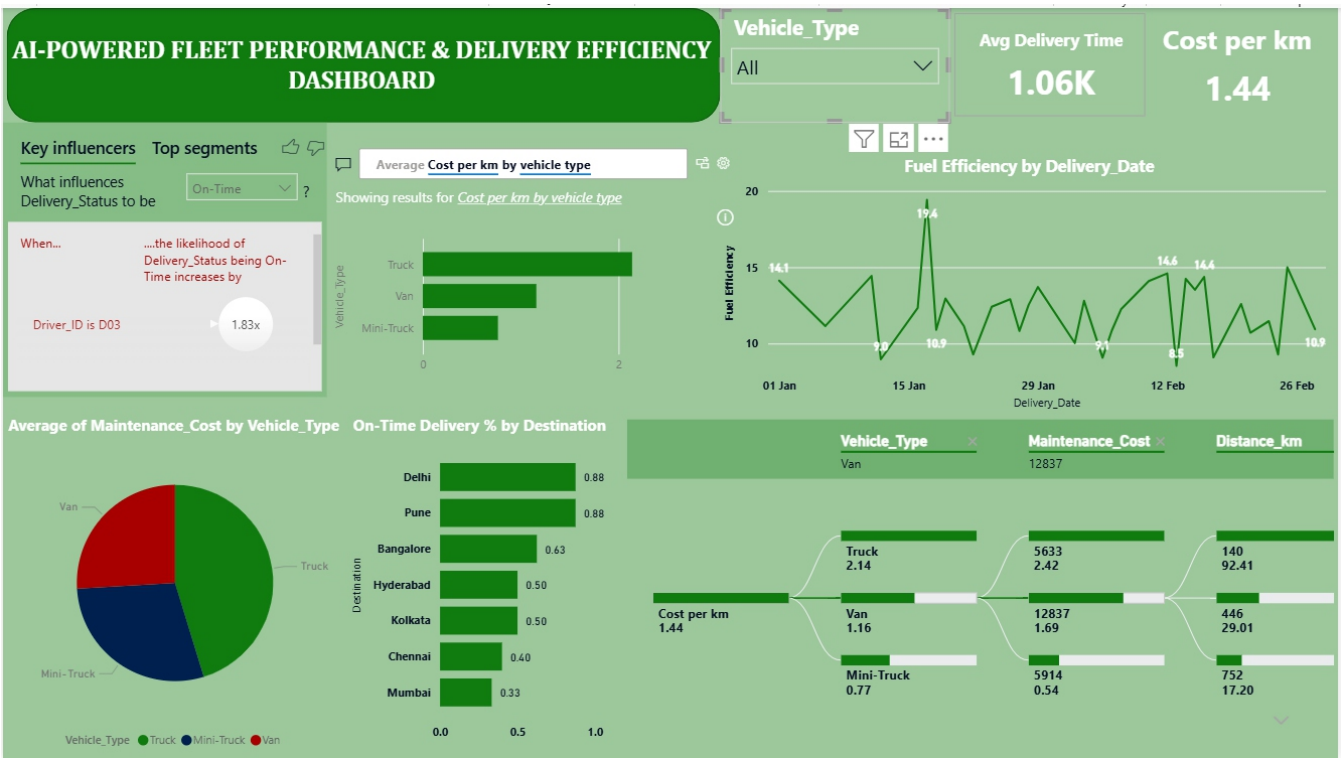




A Q&A visual was used to interactively analyze the average cost per kilometer by vehicle type using natural language queries.Line 2 (Support):This visual allows users to quickly ask business questions and receive instant insights, helping compare transportation costs across different vehicle types.



A slicer was added for vehicle type to enable interactive filtering of the dashboard. This allows users to analyze fleet performance and delivery efficiency for specific vehicle categories such as trucks and vans. Using the slicer helps compare costs, fuel efficiency, and delivery performance across different vehicle types.



The AI-powered dashboard provides clear insights into delivery performance, fuel efficiency, and transportation costs. It helps identify delay-prone routes, cost drivers, and efficient vehicle usage. By using interactive and AI-driven visuals, the dashboard supports data-driven decisions to optimize delivery routes and improve overall fleet utilization.